[INCH-POUND]
MIL-C-29600/3
22 February 1990

MILITARY SPECIFICATION SHEET

CONNECTOR, ELECTRICAL, CIRCULAR, MINIATURE, COMPOSITE, HIGH DENSITY, QUICK COUPLING, ENVIRONMENTAL RESISTANT, REMOVABLE CRIMP CONTACTS, BACKSHELL 90°, SELF LOCKING (SERIES A AND B)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the backshell described herein shall consist of this specification and the latest revision of MIL-C-29600.

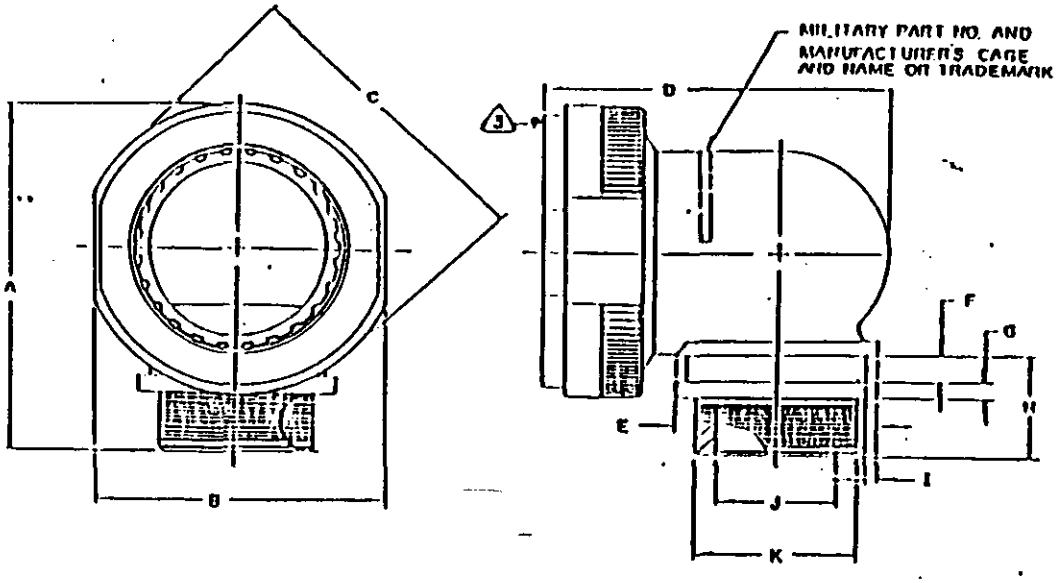


FIGURE 1. 90° backshell.

NOTES:

1. Dimensions are in inches unless otherwise specified.
2. Metric equivalents are based upon 1 inch = 25.4 mm.
3. See MIL-C-29600 for interface dimensions.
4. Method of shield termination is optional.

AMSC N/A FSC 5935

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TABLE I. Dimensions.

Shell Size	A ±.050	B Flats	C ±.015	D ±.050	E ±.015	F	G	H	I	J ±.010	K	MS9068 dash No.
08-09												-013
10-11												-015
13												-017
14-15												-019
16-17	1.750	1.375	1.450	1.625	.925	.140 .130	.100 .090	.690 .450	.060 .450	.560	.750 .743	-021
18-19												-023
20-21												-025
22-23												-027

Inches	mm	Inches	mm	Inches	mm
.002	0.058	.060	1.524	.743	18.87
.003	0.076	.090	2.286	.750	19.05
.006	0.152	.100	2.540	.925	23.50
.010	0.254	.130	3.302	1.375	34.93
.012	0.305	.140	3.556	1.450	36.83
.015	0.381	.450	11.43	1.625	41.28
.045	1.143	.560	14.22	1.750	44.45
.050	1.270	.690	17.53		

TABLE II. Qualification tests. 1/

Inspection	Requirement Paragraph	Test Paragraph
Examination of product	3.1 thru 3.5, 3.41 and 3.42	4.6.1
Life cycle	MIL-C-29600/3	MIL-C-29600/3
EMI shielding effectiveness	3.27	4.6.22
Temperature cycling	3.8	4.6.3
Vibration	3.22	4.6.17
Shock	3.23	4.6.18
Shield to shell conductivity	MIL-C-29600/3	4.6.19.3
External bending moment	3.17	4.6.12
Salt spray (2000 hours)	3.15	4.6.10.1
Accessory thread strength	3.39	4.6.35
Impact	3.38	4.6.33
Fluid immersion	3.29	4.6.24
Plating adhesion	3.28	4.6.32
Examination of product	3.1 thru 3.5, 3.41 and 3.42	4.6.1

1/ Unless otherwise specified all qualification tests shall be performed in accordance with MIL-C-29600.

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REQUIREMENTS:

Requirements shall conform to MIL-C-29600.

Dimensions: See Table I.

Design characteristics and configuration: See Figure 1.

O-ring: Use MS9068 per table.

Shield to shell conductivity: When tested as specified in MIL-C-29600, paragraph 4.6.19.3, the maximum resistance measured between the shield terminated to the connector plug and the receptacle flange shall be 20 milliohms.

Self-locking device:

(1) The self-locking devices shall provide a positive detent and be of corrosion resistant material.

(2) For testing purposes, the coupling nut shall be torqued down to 80 percent of the applicable coupling thread strength. Tolerance shall be $\pm$ 5 inch-pounds.

(3) Life cycle test: The coupling with the locking device engaged shall be rotated 60 full turns in a clockwise direction and 60 full turns counter clockwise, prior to standard qualification testing. Cycle speed shall not exceed one full turn per second.

Material: The material shall be as specified in MIL-C-29600, paragraph 3.3.2.

Finish: The finish shall be as specified in MIL-C-29600 paragraph 3.3.7.

Manufacturer's identification: All components of this backshell shall contain the manufacturer's CAGE No. and name or trademark. Marking of washers and followers is not required. All components of the final backshell assembly shall be of the same manufacturer.

Qualification: See Table II. Termination of the shield to the backshell shall be accomplished by mechanical, thermal, or electromechanical means.

Shell size ranges: Small 8-13, medium 14-19, large 20-25.

Sample preparation: Each sample shall be wired to applicable wired connectors or equivalent hardware.

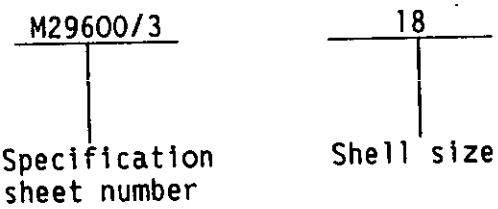
Sample size: Two samples in each shell size range and each finish.

Application: For use with MIL-C-29600 Series A and B connectors.

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REQUIREMENTS (continued):

Example of part number:



Custodians:
Army - CR
Navy - AS
Air Force - 85

Review Activities:
Army - AV
Navy - EC
Air Force - 15
DLA- ES
NASA- NA

Users:
Army - MI
Navy - SH,MC
Air force- 15

Preparing activity:
Navy - AS
(Project No. 5935-3701-15)

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